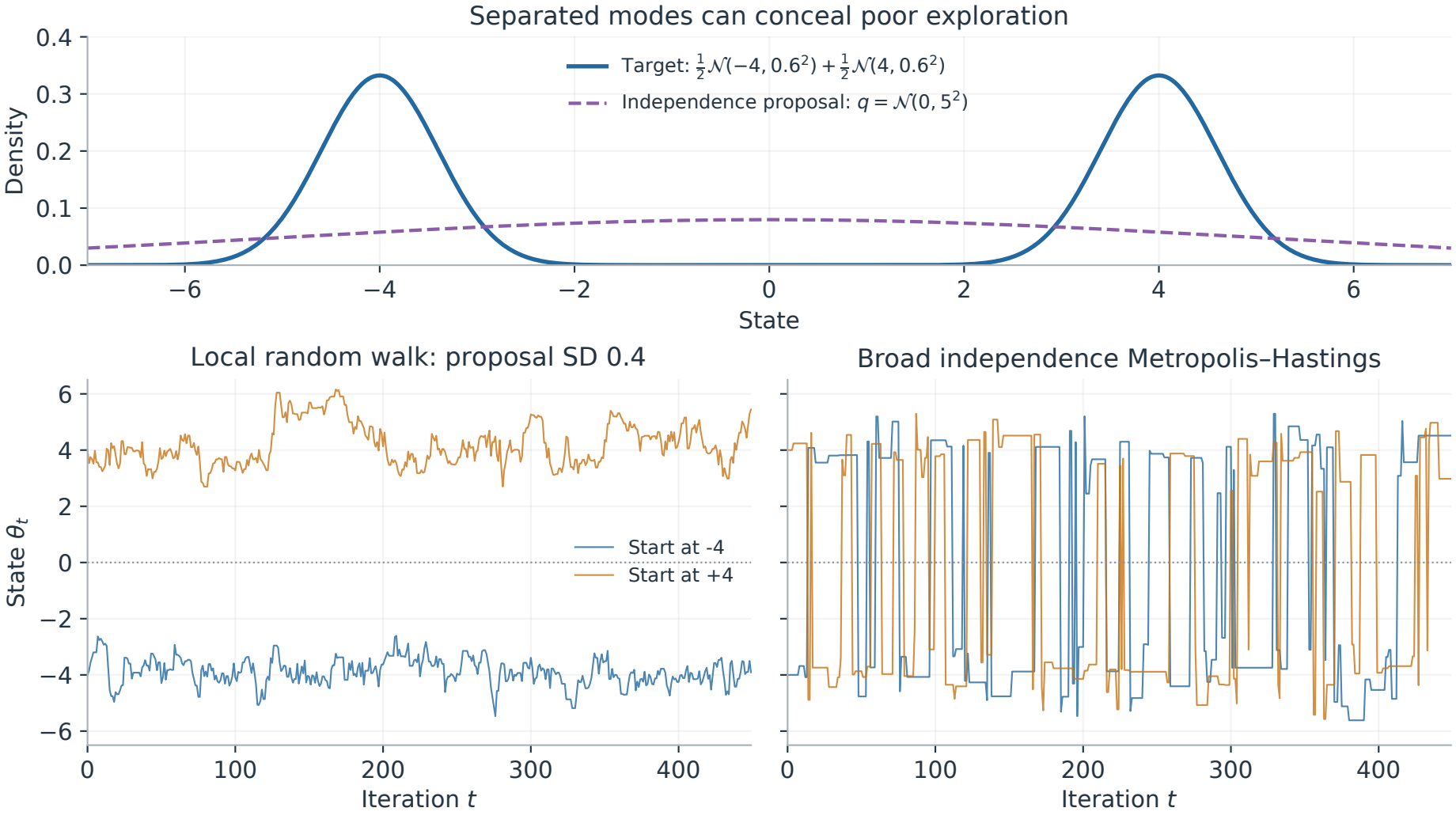
Separated modes can conceal poor exploration
Target: $\frac{1}{2}\mathcal{N}(-4, 0.6^2) + \frac{1}{2}\mathcal{N}(4, 0.6^2)$
Independence proposal: $q = \mathcal{N}(0, 5^2)$
Density
State
Local random walk: proposal SD 0.4
Broad independence Metropolis–Hastings
Start at -4
Start at +4
State θ_t
Iteration t